

Kernel

1. What is the procedure for removing and installing modules in the running kernel? Why is this procedure necessary?
2. Describe the differences between the insmod and modprobe commands.
3. Which file stores optional parameters used by kernel modules?
4. Describe the nature of a monolithic kernel and the consequences and/or advantages of using one.
5. Name the major steps required to configure, build, and install a custom kernel and its modules.

Exercises

1. Using the procedures found in Chapter 13, as well as the kernel HOWTO, configure, build, and install a custom kernel and modules. Boot the new kernel. Does your system behave normally? Are you able to boot both your original kernel and the new one?
2. Using lsmod, examine the modules loaded in your running kernel.
 1. Try removing a noncritical module, for example, rmmmod sound. Did the command fail because the module was in use?
 2. Try inserting a module, for example, modprobe fat, followed by lsmod. Did the module get inserted correctly? Remove it with rmmmod fat. Was the removal successful?
 3. What is logged in /var/log/messages during these changes?

Boot, Initialization, Shutdown, and Runlevels

1. Name and briefly describe the two parts of LILO. Which part has a configuration file and what is that file called?
2. What are the ramifications relating to new hardware when running a monolithic kernel?
3. Which three runlevels are well-defined across Linux distributions, and what actions do they perform?
4. Describe a situation that would imply the need to switch to single-user mode.
5. How can you shut down and halt a Linux system immediately using shutdown?

Exercises

1. Examine the contents of /etc/lilo.conf. How many kernel images or operating systems are configured for load by LILO? Explain the options you find in the file.
2. Install the boot loader by executing lilo. What happened?
3. Boot your system and manually specify the root filesystem using the root= keyword at the LILO prompt. What happens if you specify the wrong partition?
4. Use dmesg and less to examine boot-time messages. Compare what you find to the latest boot messages found in /var/log/messages.
5. Boot your system and use the single or 1 option to boot directly into single-user mode.

Runlevels

1. After booting to single-user mode, switch to your normal runlevel using `init n`.
 1. Does the system come up as expected?
 2. Enter `init 1` to go back to single-user mode. What daemons are still running?
2. Familiarize yourself with the contents of a few of the scripts in `/etc/init.d` (your system directories may vary).
3. Look in `rc0.d` through `rc6.d` for links to the scripts you examined. How are the scripts used? In which runlevels is the corresponding service active?
4. Shut down your system with `init 0`.
5. Shut down your system with `shutdown -h now`.

Printing

1. What does `lpd` do to handle incoming print jobs destined for empty print queues?
2. Describe the kinds of information included in `/etc/printcap`.
3. What is the function of a print filter?
4. What does the `-P` option specify to the print commands?
5. When is it useful to pipe into the standard input of `lpr` instead of simply using a filename as an argument?
6. How is the Ghostscript program used in printing to a non-PostScript printer?
7. What filter is used on a Linux system to print to remote printers on Windows clients?

Exercises

1. On a system with an existing printer, examine `/etc/printcap`. Which print filter is used for the printer? Which queue or queues are directed at the printer?
2. Check the printer status with `lpq -P printer` and `lpc status`. Print to the queue using `lpr -P printer file`.
3. Examine `/var/spool/lpd` for the spool directory of your print queue. Examine the files you find there.

Documentation

1. Describe the `PAGER` environment variable.
 1. How does it affect the `man` facility?
 2. If `PAGER` is not set, how does `man` display output?
 3. Does this environment variable affect the `info` facility?
2. In response to your query on a library function, `man` returns a page on an identically named user command. Why does this happen?
 1. How do you display the page for the function and not the command?
 2. How do you display both?
3. Name the program that displays GNU texinfo pages.

Exercises

1. Run a man inquiry as follows:

```
$ man -a -Pless mkfifo
```

There are both an mkfifo command and an mkfifo function. You'll be looking at the mkfifo command from section 1 of the manual. Note MKFIFO(1) at the top of the page.

Press q to terminate the pager program. The pager is then invoked again and displays the mkfifo function from section 3 of the manual. Note MKFIFO(3) at the top of the page.

Run the man command again, using the -Pmore option as follows:

```
$ man -a -Pmore mkfifo
```

1. What differences do you see in the output?
2. What does the -P option do?

2. Run another man inquiry as follows:

```
$ man -d ln
```

1. What output do you get from man?
2. What is the -d option?
3. Did you see information on the ln command?

3. Now examine the man configuration file:

```
$ less /etc/man.config
```

4. Notice how the contents of this file coincide with the result you received from the -d option.

Acting as a Linux helpdesk

Suppose you are a help desk technician in a mixed-systems office, and you are relatively new to Linux. A user calls your help desk with a general question about Linux system shutdown. He indicates that he's heard from Unix gurus that using the halt command could be unsafe. He also reports getting frustrated with Windows NT users who use the Ctrl-Alt-Delete key combination on his system console, which causes his Linux server to reboot. He asks for specific information on:

How to safely shut down Linux

How to allow nonsuperusers the ability to shut down cleanly

How to disable the Ctrl-Alt-Delete shutdown

Let's further assume you don't know how to answer these questions and that you have access to system documentation. Complete the following steps:

1. Use the man facility to investigate the halt command. Based on what you find there, answer the following:

1. In what section of the manual is halt located? Why?
2. Determine if it is "safe" to use halt to shut down Linux. What caused the Unix gurus to instruct the caller that using halt was not safe?
3. Determine if it would still be safe if the user uses the -n option to halt.
4. Is it appropriate to use halt on a multiuser system to which others are logged in?
5. Use man on the other commands referred to by the halt manpage in the SEE ALSO section.

2. Evaluate the other commands:

1. Which commands can be used to shut down the system in place of halt?
2. Which commands would be the most appropriate for shutting down a multiuser system?

3. From what you see in the manpages:

1. Where is the Ctrl-Alt-Delete system shutdown configured?
2. Explain how to disable it.
3. Do you need to reboot to enable the change? If so, why?
4. How can you configure the system to allow specified nonsuperusers to shut down cleanly?
5. If you use the info command, are you provided with additional information?

4. After successfully following your instructions, the user calls again. This time he is puzzled by error messages that are produced when his users attempt a clean shutdown from multiuser mode using the shutdown command without arguments.

1. Reevaluate the manpages in question. Are there any clues to common problems? (Hint: see BUGS.)
2. State the typical shutdown command to issue from multiuser mode.

Shells, Scripting, Programming, and Compiling

1. What characteristic of a bash variable changes when the variable is exported?
2. Describe the concept of shell aliases.
3. When is a shell function more suitable than a shell alias?
4. Describe the function of /etc/profile.
5. What must the author of a new script file do to the file's mode?
6. How does the shell determine what interpreter to execute when starting a script?
7. How can a shell script use return values of the commands it executes?

Exercises

1. Using bash, enter the export command and the set command. Which set of variables is a subset of the other? What is the difference between the variables reported by export and those reported by set? Finally, enter which export. Where is the export command located?
2. Examine /etc/profile. How is the default umask set? What customizations are done in the file for system users?
3. Create a simple bash script using the #!/bin/bash syntax, set the executable mode bits, and execute the shell. If it runs correctly, add errors to see the diagnostic messages. Have the script report both exported and nonexported variables. Verify that the nonexported variables do not survive the startup of the new shell.

Administrative Tasks

1. Why is it considered insecure to store encrypted passwords in /etc/passwd?
 1. What is the alternative?
 2. When the alternative is implemented, what happens to the password field in /etc/passwd?
2. What would happen to a user account if the default shell were changed to /bin/false?
3. When a new account is created with useradd -m, what files are used to populate the new home directory?
4. Compare and contrast the execution of /etc/profile and /etc/bashrc.
5. What is the complete filename for the file where most syslog messages are sent?
6. Describe the three syslog parameters: facility, level, and action.
7. Compare and contrast cron and at.

8. Is there a cron command?
9. State the format of a crontab file, describing each of the six fields.
10. What does an asterisk mean in crontab fields 1 through 5?
11. Compare and contrast the differential and incremental backup methods.
12. Why is mt usually used along with tar to implement simple backup schemes?
1. What special measures must be taken with regard to device names when using mt for multiple-volume tar backups?

Exercises

User accounts

1. Examine the /etc/passwd file on your system.
 1. Is this the only means of user authentication on your system?
 2. Are shadow passwords in use?
2. Repeat the first exercise for groups.
3. If you have an expendable system available, experiment with implementing shadow passwords.
4. Add a user with useradd, including a new home directory populated with files from /etc/skel.
5. Add a group with groupadd.
6. Use usermod to add your new user to the new group.
7. Set the new user's password using passwd.
8. Log into the new account, and use newgrp to change to the new group.
9. Delete the new group and user (including home directory) using groupdel and userdel.

User environment and variables

1. Examine the contents of /etc/skel. How similar are they to your own home directory?
2. Review the contents of /etc/profile and /etc/bashrc.

Syslog and log files

1. Add the local5 facility to your configuration as described in Chapter 18. Use logger to write to your new log file, and verify its contents. Compare your log entries with those in /var/log/messages.
2. Examine /etc/logrotate.conf. What happens after /var/log/messages is rotated?

cron and at

1. Add an entry in your personal crontab file to perform a task, such as sending you an email message. Confirm that the action occurs as expected. Experiment with the five time specifiers.
2. Schedule a command in the future with at. How is at different from cron?

Backup

1. Imagine that you have recently been made responsible for an important production system. No formalized backup procedures are in place. Backup operations that are run are not cataloged and media are scattered. Now imagine that after a holiday weekend the system has crashed due to a power failure. Upon restart, the system has severe disk errors requiring manual fsck. After repairs are complete, the system is again usable, but users complain about missing, truncated, or corrupt files. If a formalized backup procedure had been in place, would the outcome have been different?
2. If you have a tape drive available, experiment with tar, creating small tarfiles on a tape.
 1. Using the nonrewinding tape device, create multiple archives on the tape, and use mt to position among them.
 2. Verify that the various archives you create are accessible to tar.

Networking Fundamentals

1. Describe how the subnet mask affects the maximum number of hosts that can be put on a TCP/IP network.
2. Name the three default address classes and the subnet masks associated with them.
3. The UDP protocol is said to be connectionless. Describe this concept and its consequences on applications that use UDP.
4. What user command is frequently used to send ICMP messages to remote hosts to verify those hosts' functionality?
5. Describe the contents and use of /etc/hosts.
6. In what configuration file are DNS servers listed? What is intended if the local loopback address is included there on a workstation?
7. Name two modes of the netstat command and the program's output in each case.
8. Describe why the route command is needed for a single interface on a nonrouting workstation.
9. How does traceroute determine the identities of intermediate gateways?
10. Describe the advantages and consequences of implementing DHCP.
11. What utility is used to configure and dial a modem prior to the creation of a PPP connection?
12. What are the four authentication modes commonly used during PPP negotiations?

Exercises

1. Examine your system's TCP/IP configuration using ifconfig eth0 or a similar command for your network interface. Are you using DHCP? What type of subnet are you running with? Is it a class A, B, or C address? Are you using a private address? Experiment with taking the interface offline using ifconfig eth0 down and ifconfig eth0 up.
2. Examine the contents of /etc/services.

3. Use the `dig` command to locate information from DNS servers about a domain name.
4. Examine your `/etc/hosts` file. How much name resolution is accomplished in this file manually?
5. Examine your `/etc/resolv.conf` file. How many DNS servers do you have available?
6. Execute `netstat -r`. How many routes are reported? What are the routes to the local network and interface for?
7. Use `traceroute` to examine the route to a favorite web site.
8. Using a standard modem (not a Winmodem), use `minicom` to connect to the modem and verify that it responds to the AT command.
9. Execute a manual PPP connection as described in Chapter 19. Does your modem successfully connect to your ISP? Examine `/var/log/messages` for information on the PPP session.

Networking Services

1. Describe the function of `inetd` and how it is configured.
2. Describe the TCP wrappers security tool and how it can enhance security on a networked system.
3. Describe the use of mail aliases and how to enable new aliases.
4. What files are used to configure the Apache web server?
5. How does the administrator share a directory using NFS?
6. How does the administrator mount a directory shared by a remote NFS server?
7. What is the function of `nmbd`?
8. Which file is used to configure `smbd`?
9. Describe the function and location of the resolver.
10. Name two programs that can be used to do queries against an NFS server.
11. Describe in general terms the main difference between BIND v4 and BIND v8 configuration files.

Exercises

1. Examine your `xinetd.d` directory. Are `telnetd` and `ftpd` enabled? If so, is TCP wrappers (`tcpd`) configured for them?
2. Enable TCP wrappers and place `ALL:ALL` in `/etc/hosts.deny`, then attempt a Telnet session. Is the inbound request ignored as expected?
3. Familiarize yourself with `/etc/sendmail.cf` and `/etc/aliases`. Run `newaliases`.
4. Familiarize yourself with Apache configuration files in `/etc/httpd/html/*.conf` (or your directory location).
5. Examine your `/etc/exports` file, if it exists. Are you sharing NFS volumes?
6. Examine your `/etc/smb.conf` file. Are you sharing Samba printers or volumes?
7. Examine your resolver configuration in `/etc/resolv.conf`, `/etc/hosts.conf`, and `/etc/nsswitch.conf`. Is the local `/etc/hosts` file consulted prior to DNS servers?
8. By examining `/etc/named.conf` and noting format details, determine what general release of BIND you are using.

Security

1. What daemon is associated with the control files `/etc/hosts.allow` and `/etc/hosts.deny`.
2. In general terms, describe a method to locate `setuid` programs in the local filesystem. Why might an administrator do this routinely?
3. What is the function of the `md5sum` utility?
4. Why might a user run `ssh` instead of `telnet`?
5. Describe shadow passwords and the file where the passwords are stored.

Exercises

1. Use `find` as described in Chapter 21 to locate `setuid` files. Is the list larger than you expected? Are the entries on your list justifiably `setuid` programs?
2. Use the `md5sum` utility on a binary program file and examine its output.